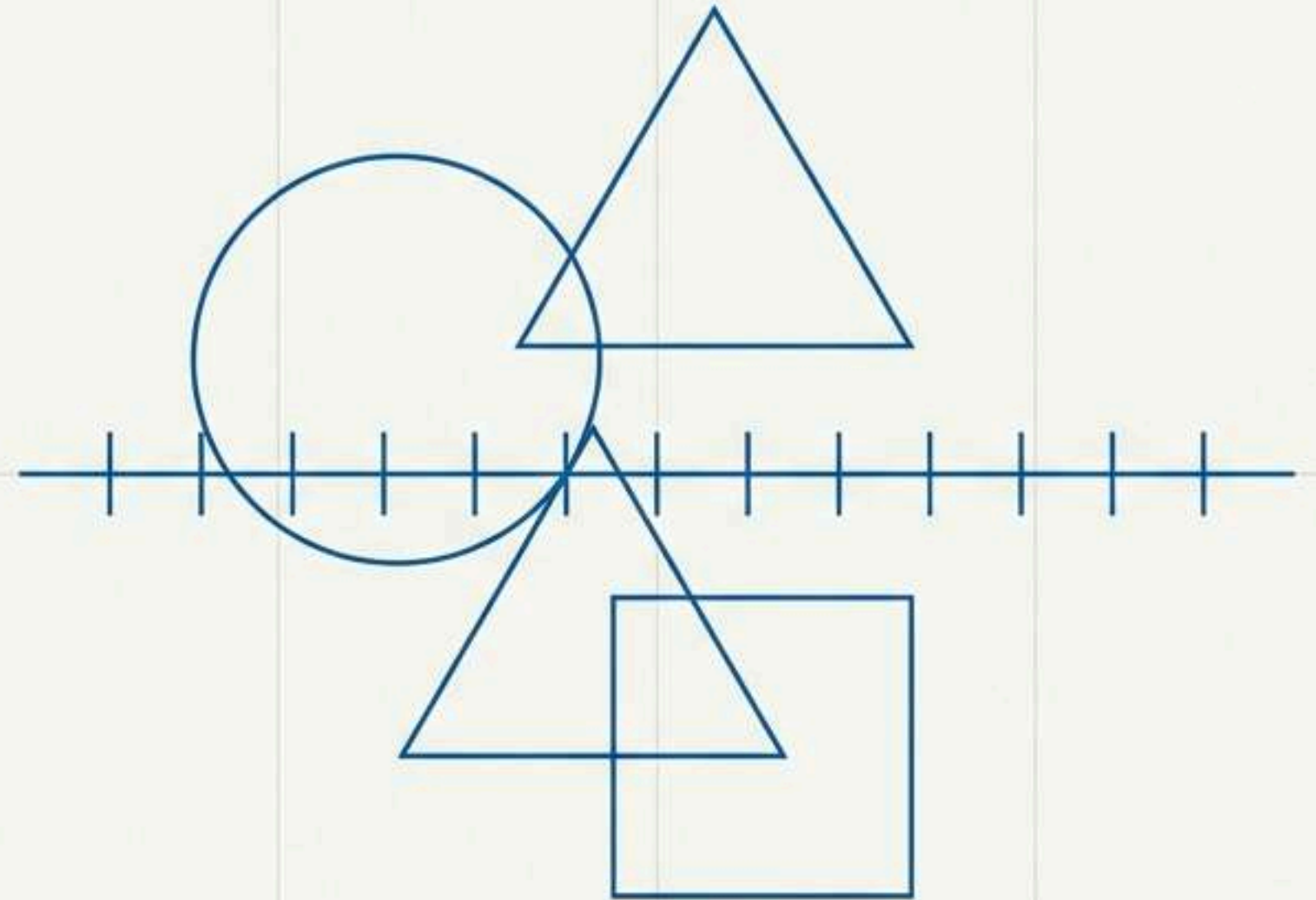


# Real Numbers

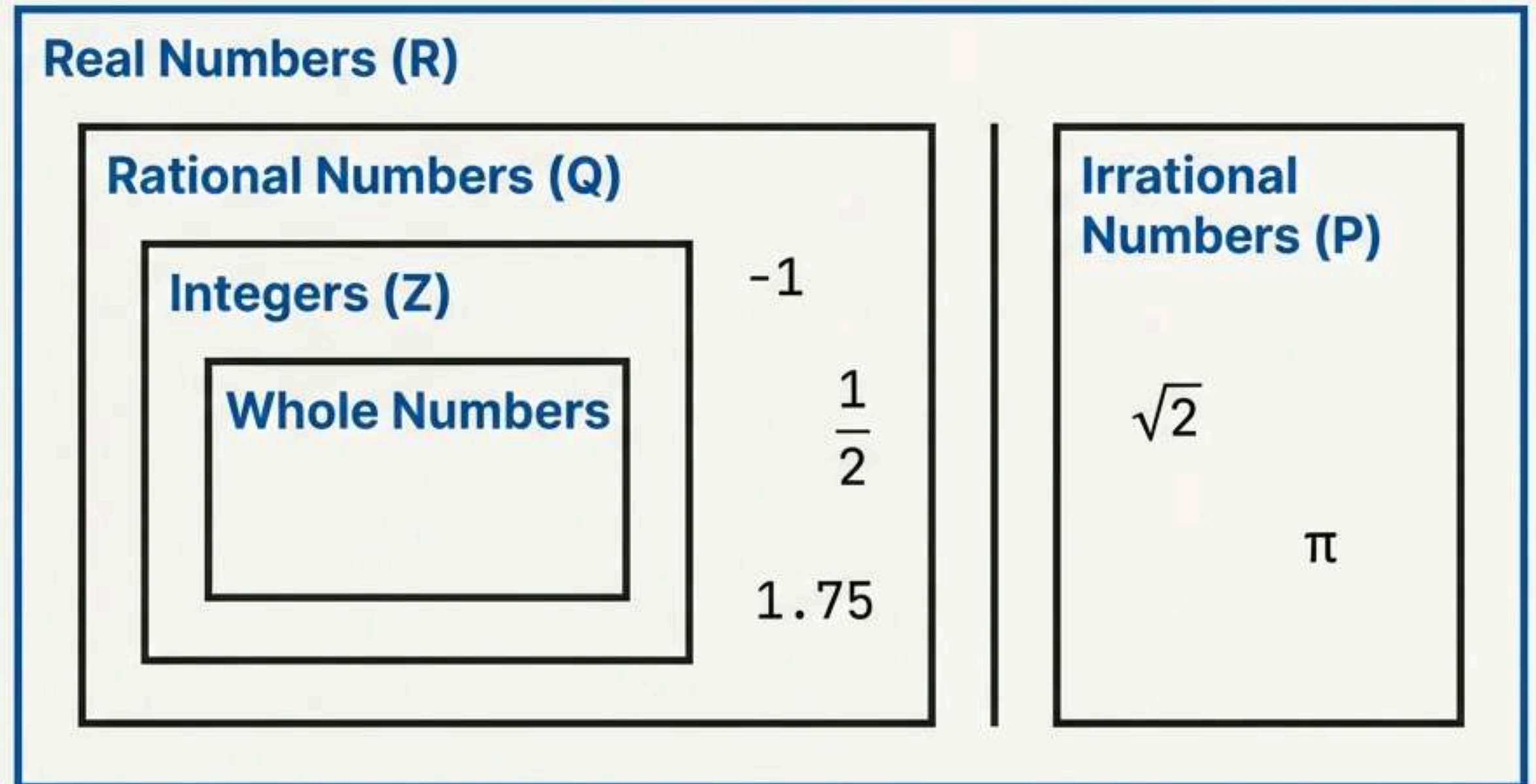
The Foundation of Mathematics

A Comprehensive Architectural Breakdown  
of CBSE Class 10 Maths, Chapter 1



# The Landscape of Real Numbers

Real numbers encompass the broad spectrum of values found in the physical world—from counting integers to measuring continuous quantities. They exclude only complex numbers.



Every real number corresponds to a unique point on the number line.

# The First Law: Euclid's Division Lemma

Formalising the arithmetic of division.

$$\begin{array}{ccccccc} a & = & b & q & + & r & \\ \text{Dividend} & & \text{Divisor} & \text{Quotient} & & \text{Remainder} & \end{array}$$

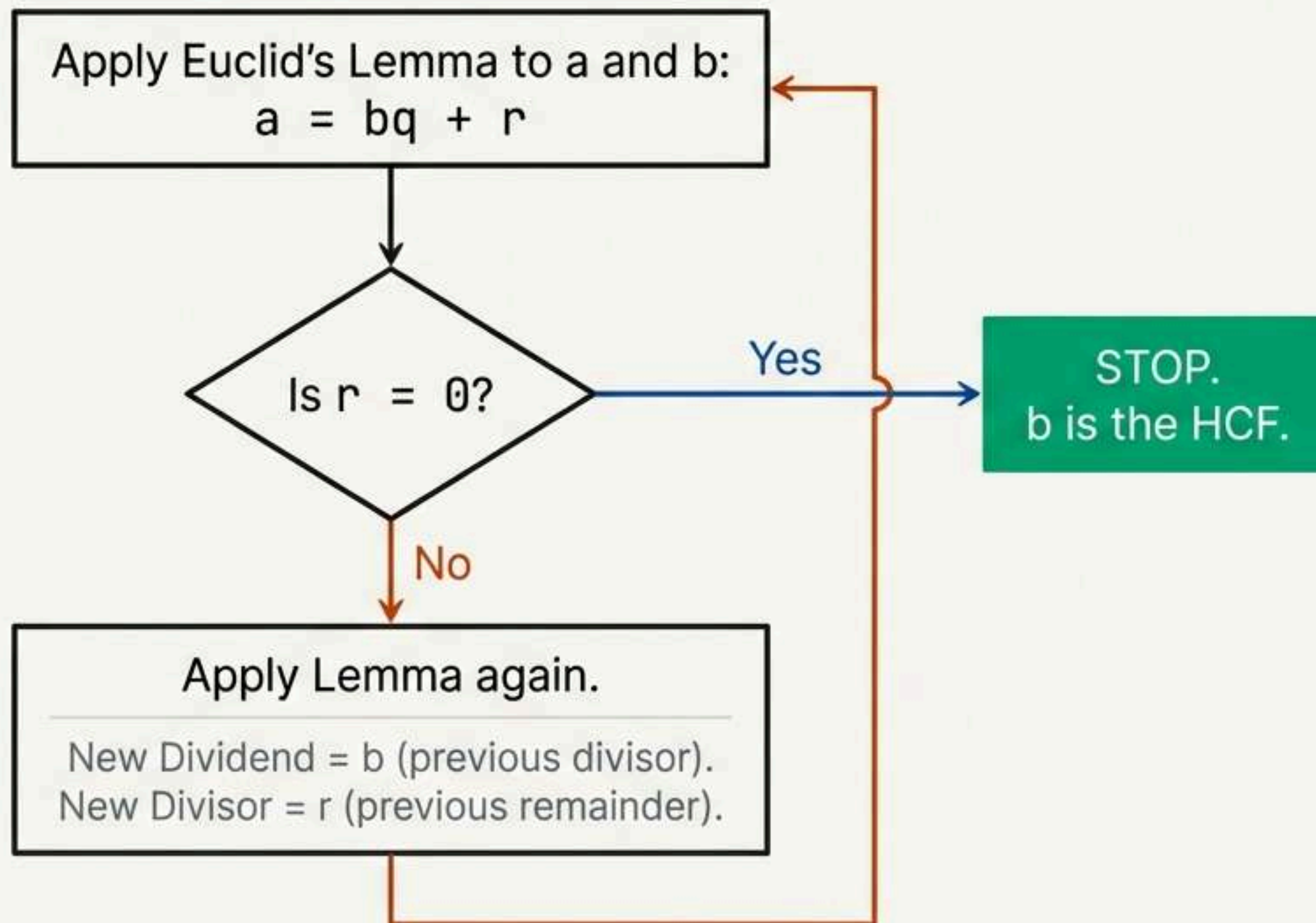
Condition:  $0 \leq r < b$

Given two positive integers  $a$  and  $b$ , there exist unique integers  $q$  and  $r$  satisfying this equation.  
Simply put: The remainder is always non-negative and smaller than the divisor.

# The Tool: Euclid's Division Algorithm

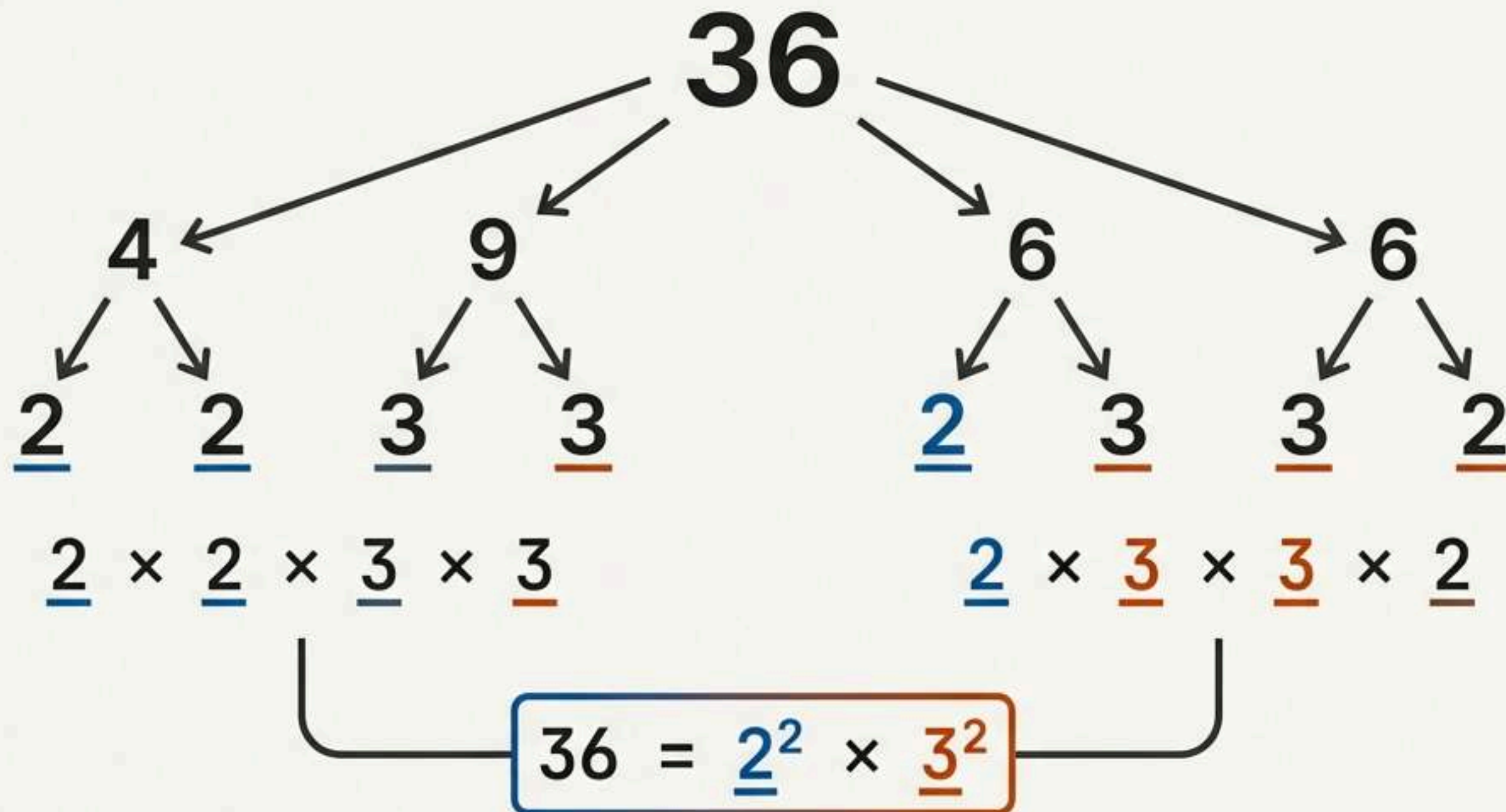
A repeatable process to find the Highest Common Factor (HCF).

Use this algorithm when dealing with large numbers where prime factorisation is difficult.



# The Second Law: Fundamental Theorem of Arithmetic

Every composite number can be uniquely expressed as a product of primes. This factorisation is unique, apart from the order in which the prime factors occur.



# Method 1: Finding the Highest Common Factor (HCF)

The greatest number that divides each given number exactly.

## Technique A: Prime Factorisation

Best for: Small Numbers

Example: 20 and 24

$$20 = 2^2 \times 5$$

$$24 = 2^3 \times 3$$

Rule: Smallest power of common primes.

Calculation:  $2^2 = 4$

## Technique B: Euclid's Algorithm

Best for: Large Numbers

Example: 18 and 30

$$30 = 18(1) + 12$$

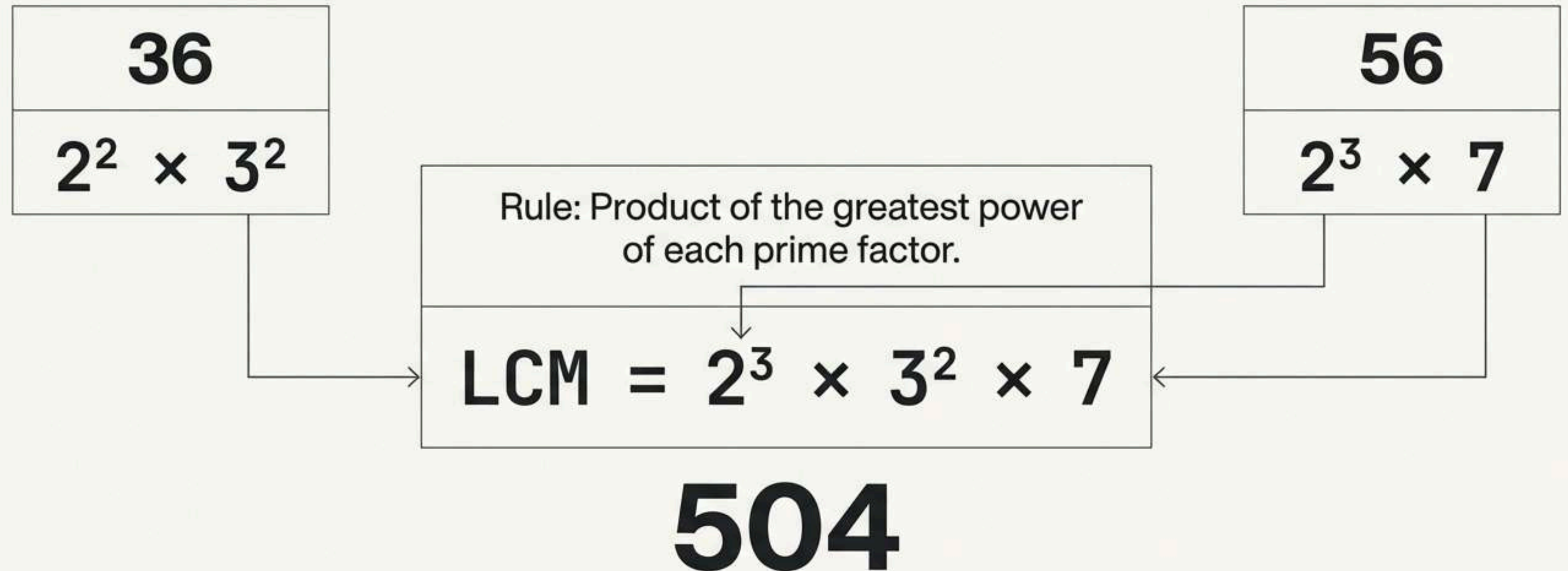
$$18 = 12(1) + 6$$

$$12 = 6(2) + 0$$

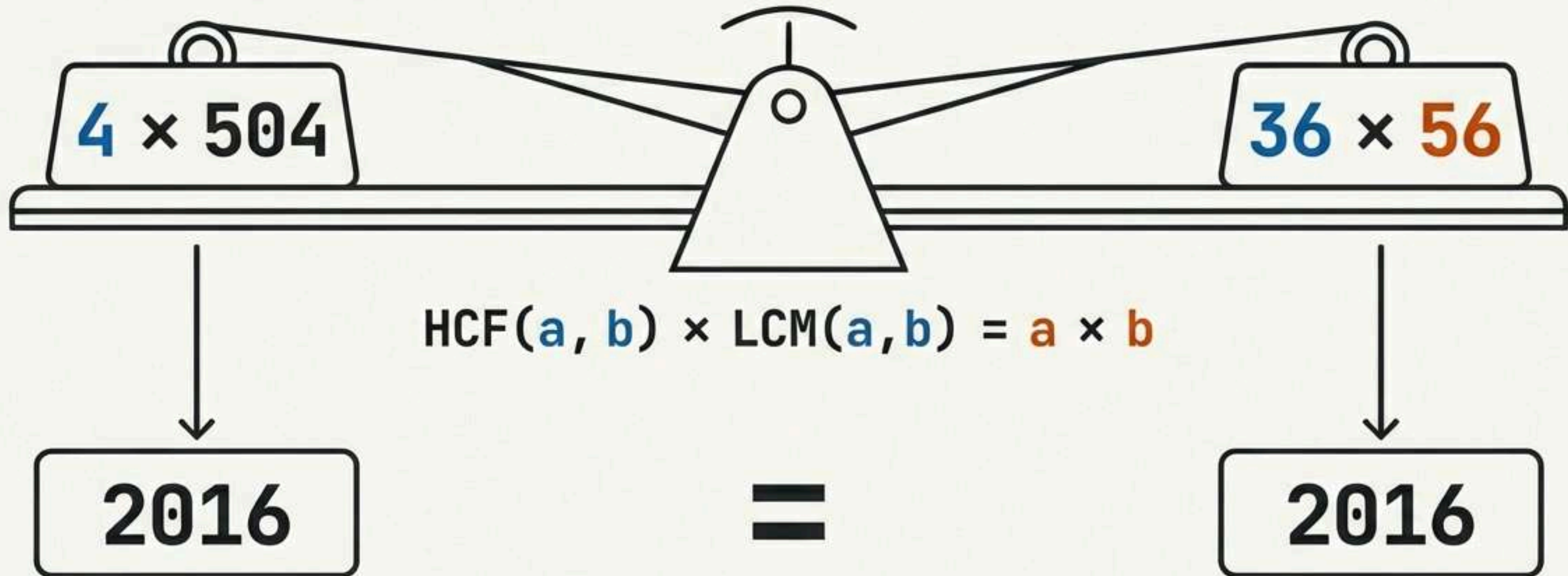
HCF is the final divisor: 6

# Method 2: Finding the Least Common Multiple (LCM)

The smallest common multiple of two or more numbers.



# The Golden Relationship



Caution: This relationship holds true for two numbers only. It does not apply to sets of three or more numbers ( $a \times b \times c \neq \text{HCF} \times \text{LCM}$ ).

# Applied Mathematics: The 'Remainder' Problem

Find the largest number that divides 70 and 125, leaving remainders 5 and 8 respectively.

1. Normalise

$$\begin{array}{rcl} 70 & - & 5 = 65 \\ 125 & - & 8 = 117 \end{array}$$



2. Factorise

$$\begin{array}{rcl} 65 & = & 5 \times 13 \\ 117 & = & 3^2 \times 13 \end{array}$$



3. Identify HCF

$$\begin{array}{rcl} 65 & = & 5 \times 13 \\ 117 & = & 3^2 \times 13 \end{array}$$

**Answer: 13**

# Applied Mathematics: The “Relationship” Problem

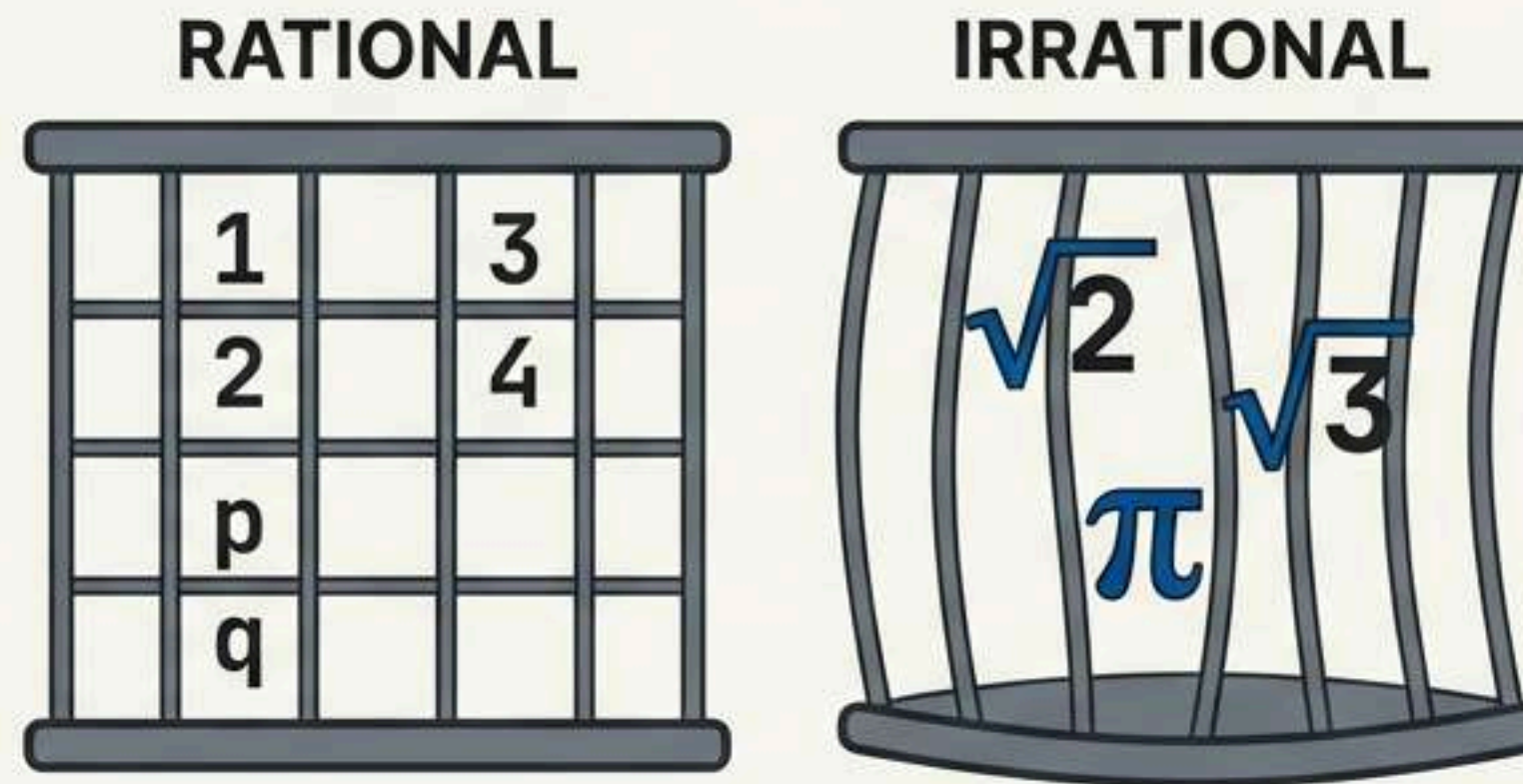
Find the LCM of 306 and 657, given that  $\text{HCF}(306, 657) = 9$ .

|       |                                                |
|-------|------------------------------------------------|
| LCM = | $\frac{\text{Product of Numbers}}{\text{HCF}}$ |
| LCM = | $\frac{306 \times 657}{9}$                     |
| LCM = | $\frac{201,042}{9}$                            |

$$\text{LCM} = 22,338$$

# The Mystery of Irrationals

Irrationals are numbers that cannot be written in the form  $p/q$  (where  $q \neq 0$ ). They have non-terminating, non-recurring decimal expansions.



## Proof by Contradiction

To prove a number is irrational, we assume the opposite (that it IS rational) and demonstrate that this assumption leads to a logical impossibility.

# The Proof: Irrationality of $1/\sqrt{2}$

## The Assumption

Assume  $1/\sqrt{2}$  is rational.

$$\frac{1}{\sqrt{2}} = \frac{p}{q} \quad (\text{where } p, q \text{ are co-prime integers}).$$

Invert the equation.  $\rightarrow \frac{q}{p} = \sqrt{2}$

## The Contradiction

**LHS**

Since  $p$  and  $q$  are integers,  $q/p$  is Rational.

**RHS**

We know  $\sqrt{2}$  is Irrational.

**Rational  $\neq$  Irrational**

The assumption leads to a contradiction.  
Therefore,  $1/\sqrt{2}$  must be **Irrational**.

# The Mathematical Toolkit

## Euclid's Division Lemma

$$a = bq + r$$

$$0 \leq r < b$$

## Fundamental Theorem

Composite Number = Unique Product of Primes

Order of factors does not matter.

## Relationship Formula

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

Valid for 2 numbers only. (Slate Grey (#6C757D))

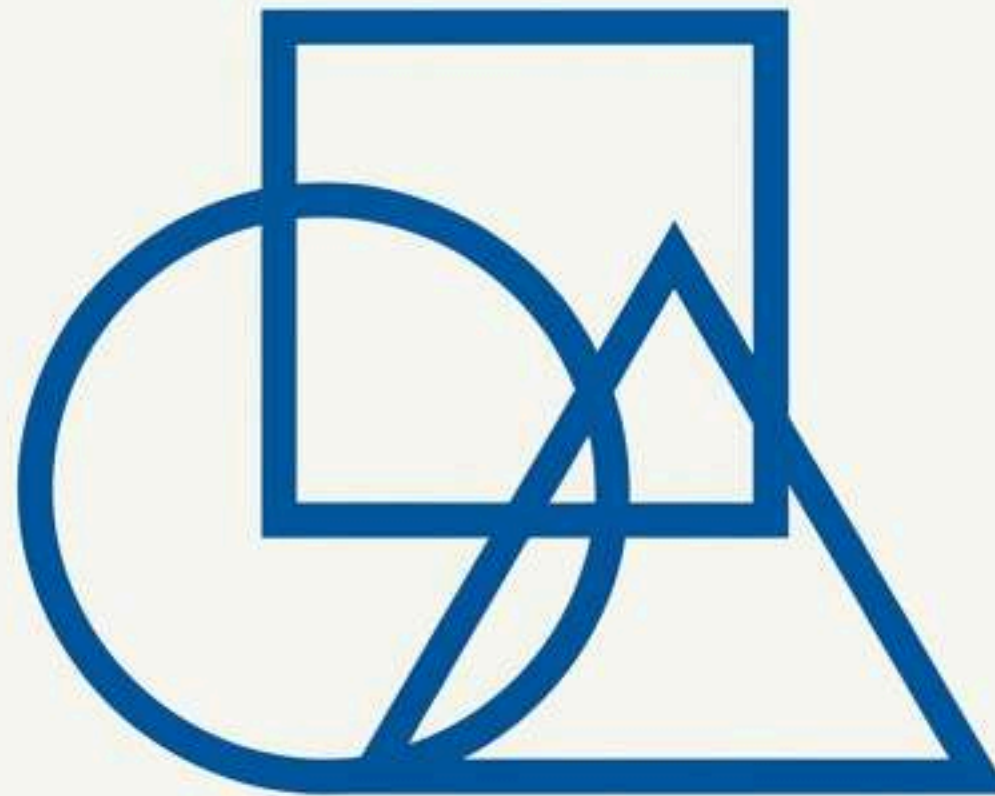
## Number Types

Rational: Can be written as  $p/q$ .

Irrational: Cannot be written as  $p/q$ .

# Final Thoughts

Real numbers are the tools we use to measure the world. Mastery of algorithms, HCF, LCM, and these fundamental theorems builds the logic needed for all advanced mathematics.



End of Chapter 1.